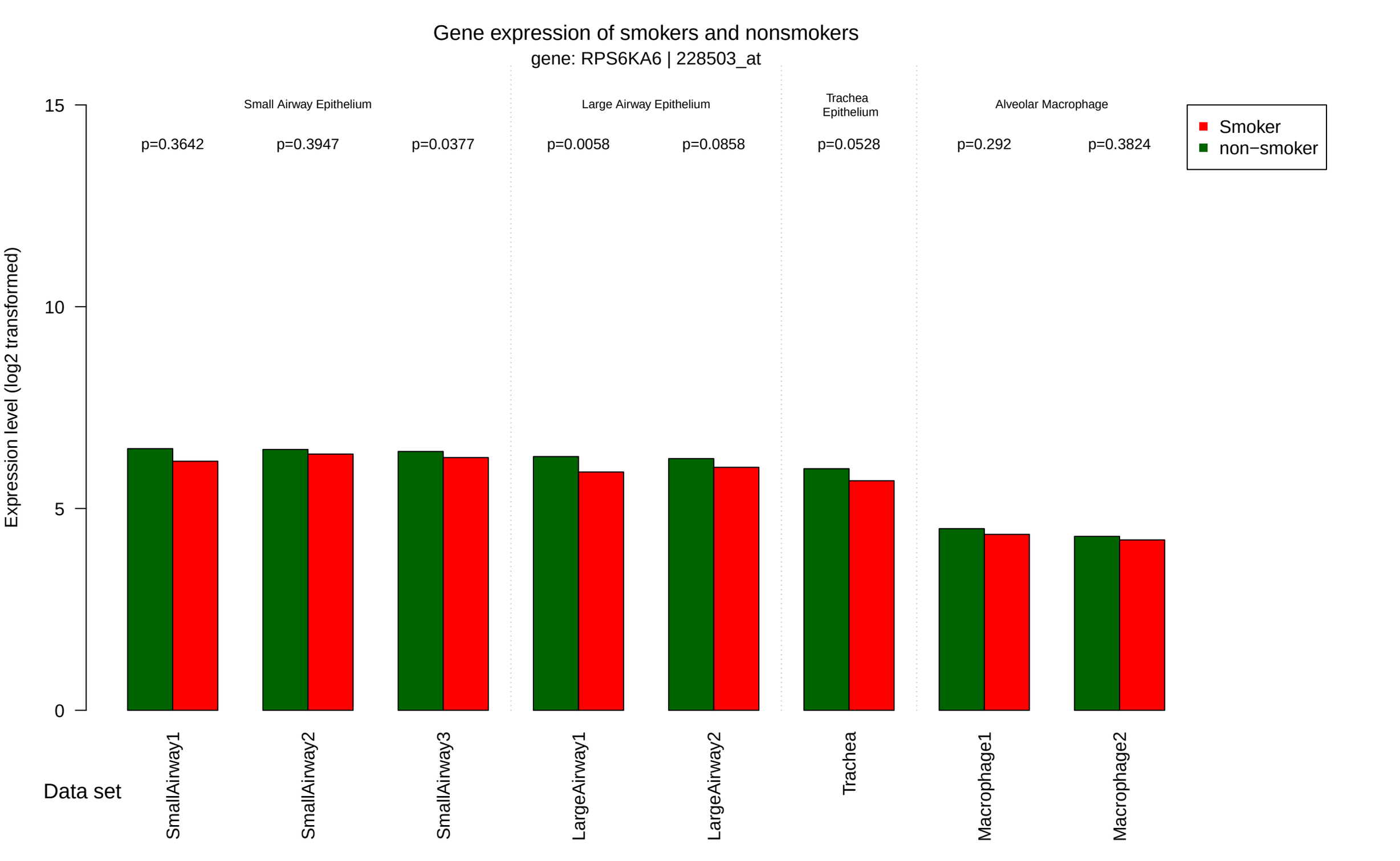
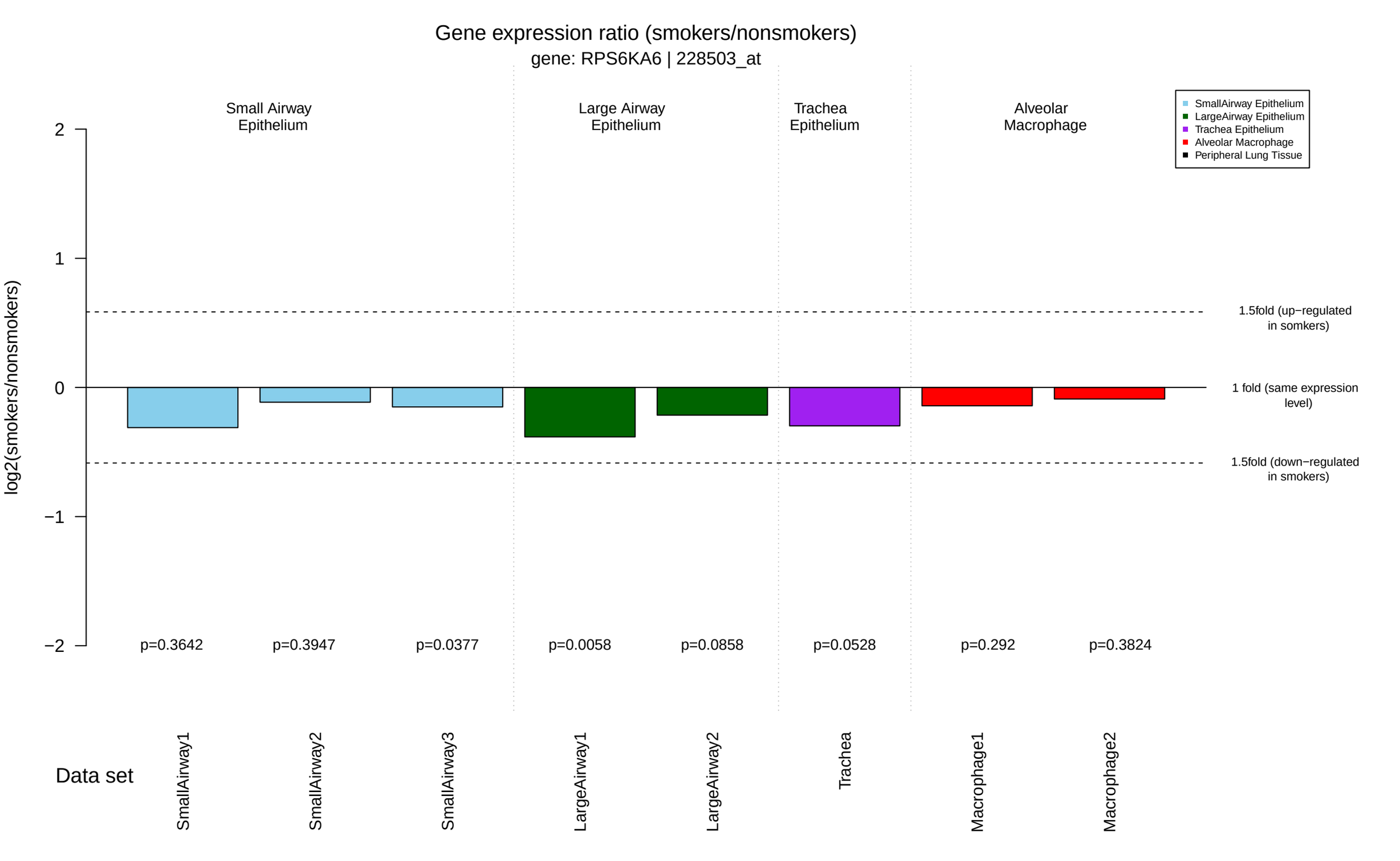
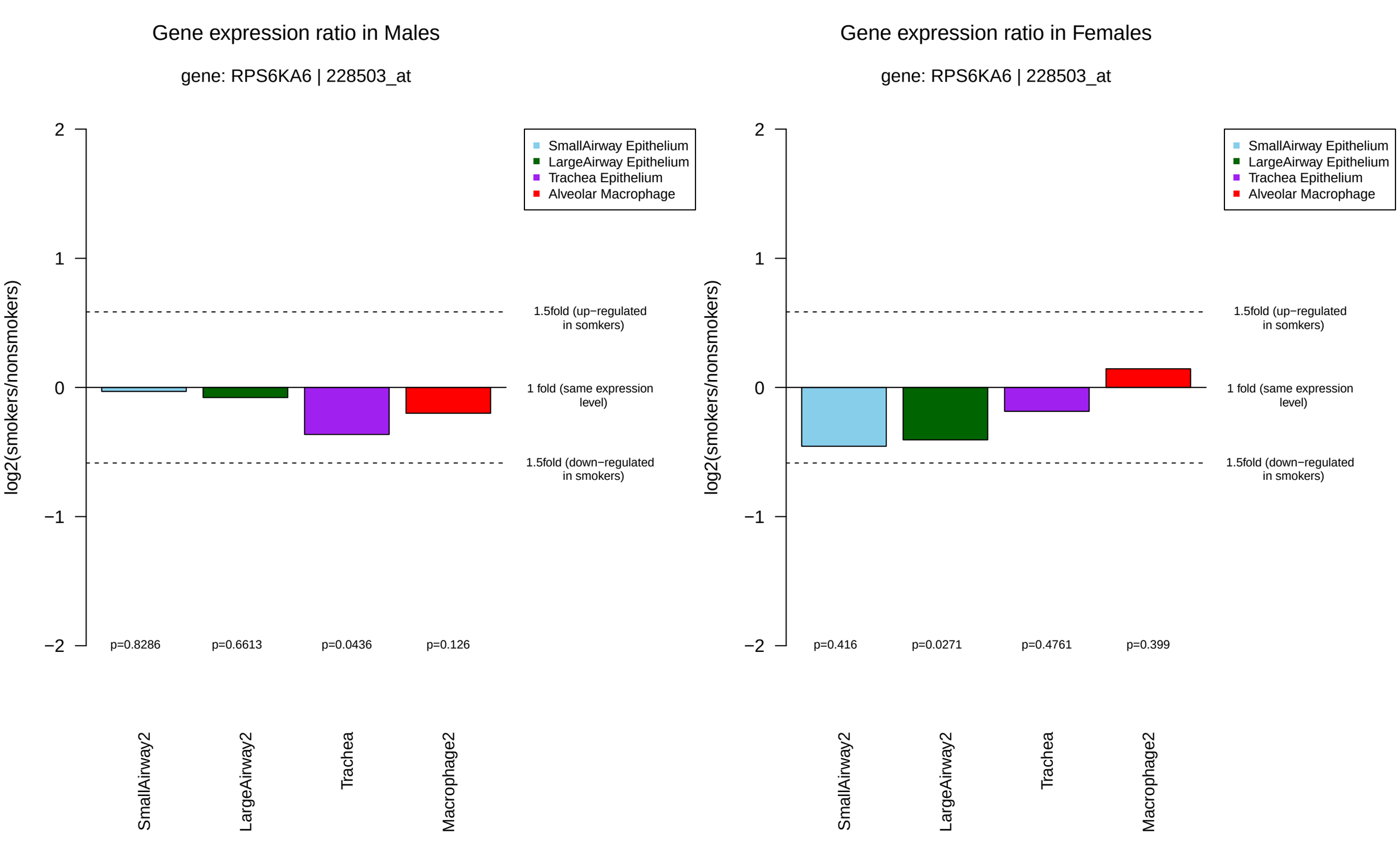
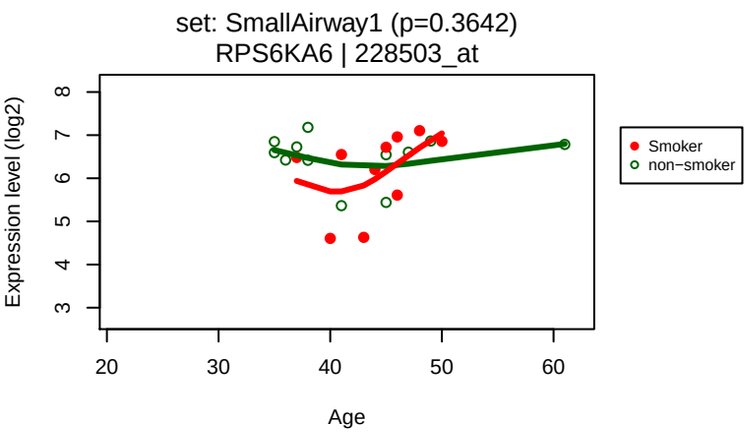
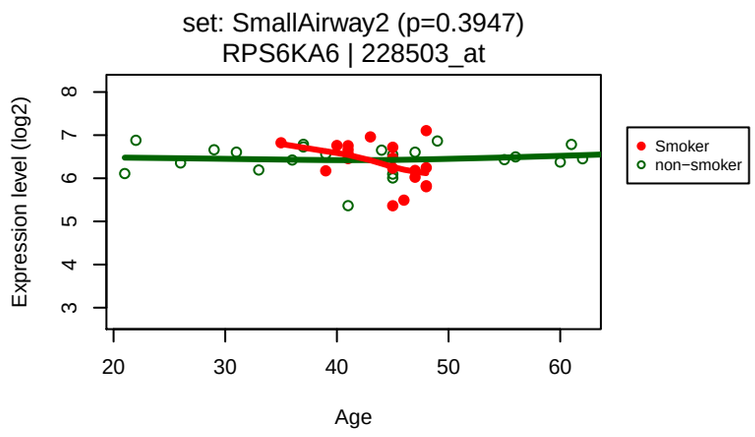
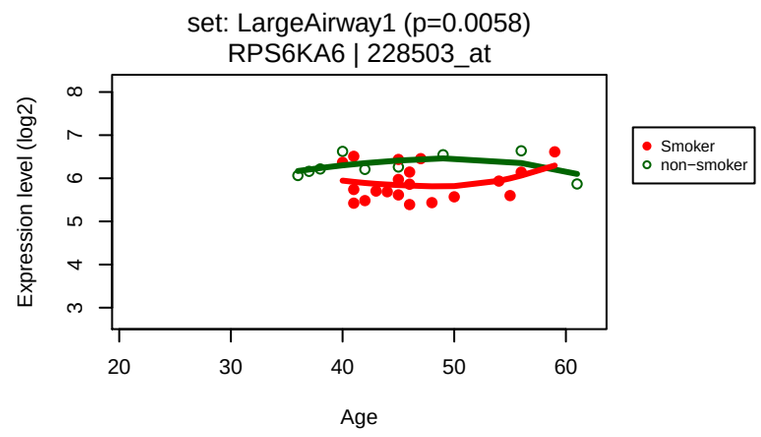

Gene expression of smokers and nonsmokers
gene: RPS6KA6 | 228503_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3642
p=0.3947
p=0.0377
p=0.0058
p=0.0858
p=0.0528
p=0.292
p=0.3824
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

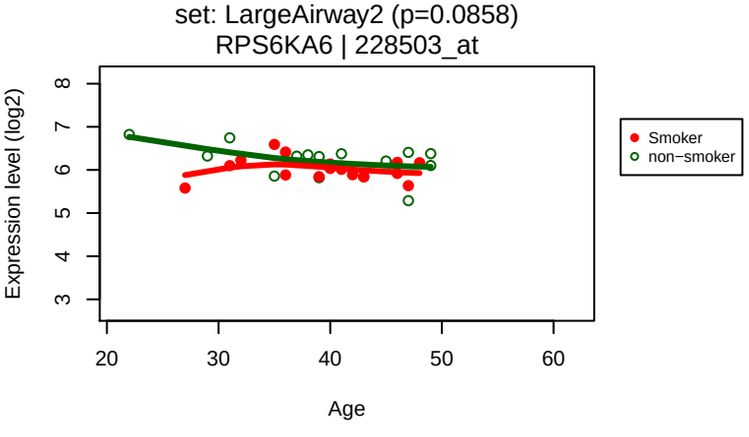

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA6 | 228503_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3642
p=0.3947
p=0.0377
p=0.0058
p=0.0858
p=0.0528
p=0.292
p=0.3824
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

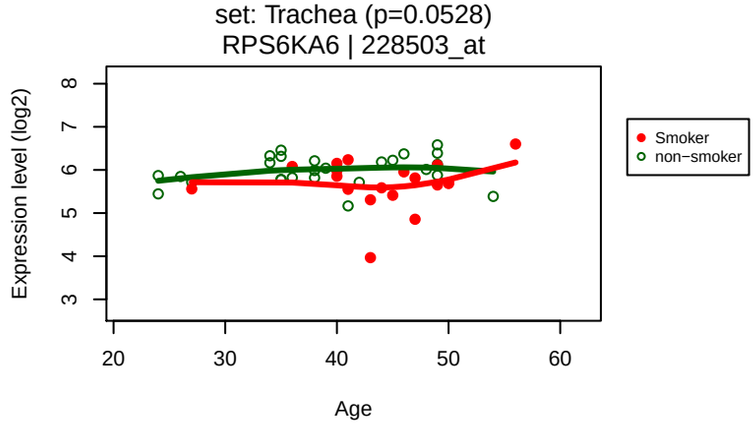

Gene expression ratio in Males
gene: RPS6KA6 | 228503_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8286
p=0.6613
p=0.0436
p=0.126
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA6 | 228503_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.416
p=0.0271
p=0.4761
p=0.399
SmallAirway2
LargeAirway2
Trachea
Macrophage2

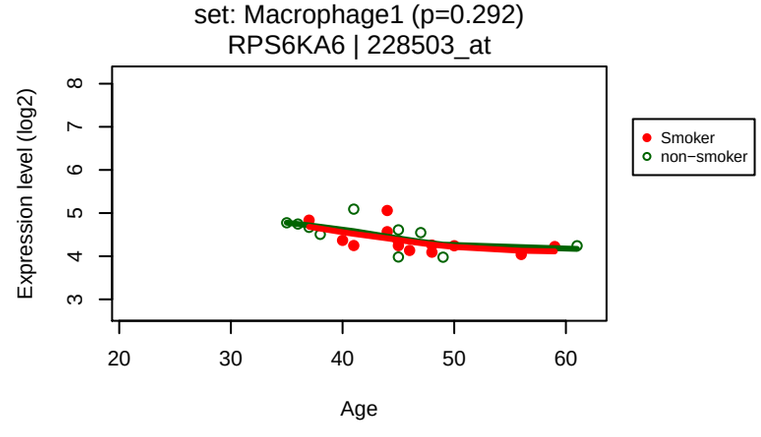

set: SmallAirway1 (p=0.3642)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker

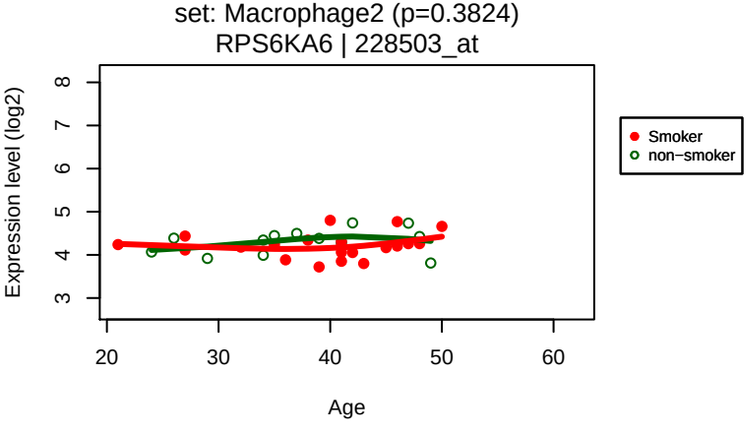

set: SmallAirway2 (p=0.3947)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker

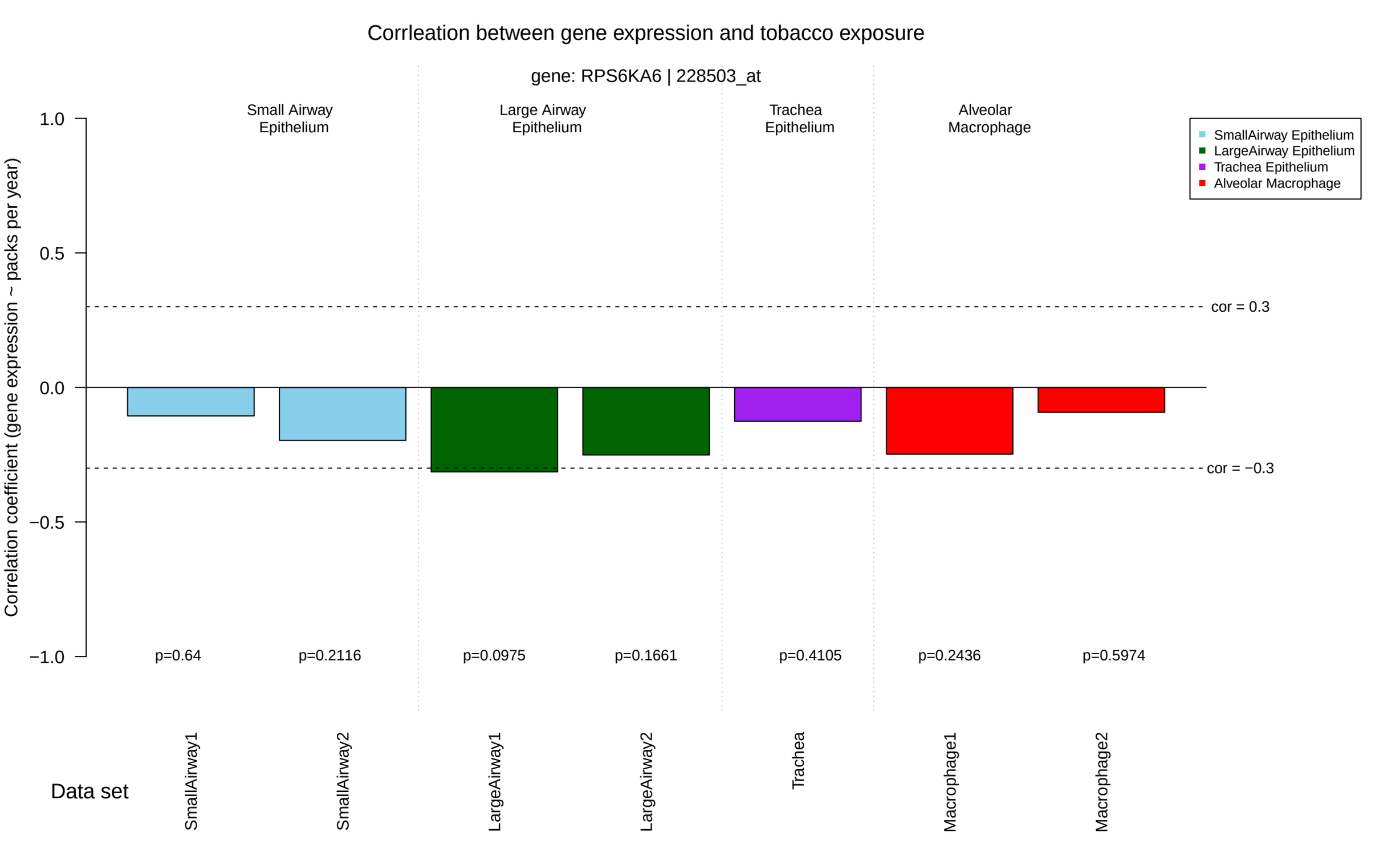

set: LargeAirway1 (p=0.0058)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0858)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0528)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.292)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3824)
RPS6KA6 | 228503_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA6 | 228503_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.64
p=0.2116
p=0.0975
p=0.1661
p=0.4105
p=0.2436
p=0.5974
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2